

Software Requirements Specification for Rajmudra Fintech Platform

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Revision History

Version	Date	Author	Description
1.0	April 2026	Platform Owner	Draft — Pending Review

Chapter 1

Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional, non-functional, and technical requirements for the Rajmudra Fintech web application — a full-stack MERN (MongoDB, Express.js, React.js, Node.js) trading education platform hosted at rajmudrafiintech.com. The platform enables student enrolment, Cashfree-powered payments, Google Meet live classes, YouTube-embedded lecture delivery, user dashboards, and a multi-tier admin system.

1.2 Document Conventions

- **FR-xx**: Functional Requirement
- **NFR-xx**: Non-Functional Requirement
- **UC-xx**: Use Case
- **UI-xx**: UI/UX Specification
- All monetary values are in Indian Rupees (INR) inclusive of GST.
- Priority levels: **Critical** $\downarrow$ **High** $\downarrow$ **Medium** $\downarrow$ **Low**

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- **Developers (MERN Stack)**: To understand the technical architecture, API specifications, and database schema.
- **Project Managers**: For timeline estimation and milestone tracking (Section 10).
- **QA Testers**: To validate acceptance criteria (Section 11).
- **Super Admin (Owner)**: To review the overall scope, security constraints, and business rules.

- **Stakeholders (Mentors):** To review user roles and admin capabilities.

Readers should start with Chapter 2 (Overall Description) and Chapter 4 (System Features) for a functional overview, then refer to specific technical sections as needed.

1.4 Product Scope

The Rajmudra Fintech platform serves as a premium Forex trading education system founded and mentored by Vishal Tompe Patil and Niles Thore Patil. It replaces the existing static HTML website with a dynamic, authenticated, payment-enabled full-stack application. The scope includes rebuilding the existing brand identity (dark luxury theme: black #080808, gold #C9A84C) and adding:

- User authentication and role-based access control (Student, Admin, Super Admin).
- Cashfree payment gateway integration for course enrolment (INR 39,999).
- Student dashboard with course phases, YouTube lectures, and Google Meet live sessions.
- Admin dashboards for student/content management and support tickets.
- Super Admin dashboard for revenue analytics, audit logs, and platform settings.

1.5 References

- Cashfree Payment Gateway API Docs: <https://docs.cashfree.com/>
- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- Existing Static Website: <https://rajmudrafinetech.com> (Design identity reference).

Chapter 2

Overall Description

2.1 Product Perspective

This product is a new, self-contained web application that replaces a static HTML website. It is a standalone system but relies on several external software interfaces:

- **Cashfree Payment Gateway:** For processing payments.
- **Google Meet:** For live class link delivery (Phase 1).
- **YouTube:** For hosting unlisted lecture videos.
- **WhatsApp Business API:** For chat integrations.
- **Email Service (SendGrid):** For transactional emails.

The system does not currently integrate with legacy databases or internal ERP systems.

2.2 Product Functions

A high-level summary of the major functions:

- **Public Website:** Preserves existing static content with React components; includes animated market ticker, hero section, syllabus, and mentor bios.
- **Authentication:** Student registration, email verification, login, JWT-based session management, and password reset.
- **Payment Processing:** Cashfree checkout, webhook handling, invoice generation (PDF), and refund management (manual).
- **Student Dashboard:** Personalized overview, sequential course phases, YouTube video embedding, Google Meet joining, support tickets, and trade audit submission.
- **Admin Dashboard:** Student management, content upload, live session scheduling, announcement posting, and ticket resolution.
- **Super Admin Dashboard:** Full user management, revenue reporting, admin activity audit logs, and platform configuration.

2.3 User Classes and Characteristics

User Class	Characteristics	Permissions/Scope
Super Admin	Platform owner. High technical expertise.	Full system access, user role management, revenue analytics, audit logs, platform settings.
Admin (Mentor 1 & 2)	Vishal Tompe Patil and Niles Thore Patil. Moderate technical expertise.	Manage students, course content, live sessions, announcements, support tickets, trade audits.
Student	Enrolled paying users. Varying technical skill.	Own dashboard, purchased course access, live session joining, support tickets, trade audit submission.

2.4 Operating Environment

- **Web Browser:** Modern browsers (Chrome, Firefox, Safari, Edge) with JavaScript enabled. Mobile responsive.
- **Server Environment:** Node.js (v18+), Express.js, MongoDB Atlas (Cloud).
- **Network:** HTTPS-enabled, supports 4G/5G and broadband connections.
- **Time Zone:** All timestamps and live sessions are in Indian Standard Time (IST, UTC+5:30).

2.5 Design and Implementation Constraints

- **Technology Stack:** Mandatory use of MERN (MongoDB, Express, React, Node.js) with Vite as the build tool.
- **Styling:** Must strictly adhere to the existing brand design tokens (colors, fonts, polygon clip-paths). No third-party UI libraries that override core styles.
- **Authentication:** JWT must be stored in `httpOnly` cookies (not `localStorage`) for security.
- **Payment:** Only Cashfree Payment Gateway; INR currency only.
- **Video Delivery:** YouTube unlisted embed only (no direct video hosting to save bandwidth).
- **Regulatory:** No refund policy (per T&C); admin exceptions only.

2.6 User Documentation

- **Student User Manual:** PDF guide on accessing dashboard, joining live classes, and submitting trade audits.
- **Admin Quick Start Guide:** Instructions for scheduling sessions, managing students, and replying to tickets.
- **In-app Tooltips:** Contextual help icons on dashboard for new users.

2.7 Assumptions and Dependencies

- **Assumptions:**
 - Students have a valid email address and WhatsApp number.
 - Students have a stable internet connection for live classes.
 - Admins will generate Google Meet links manually before scheduling.
- **Dependencies:**
 - Cashfree API availability for payment processing.
 - YouTube and Google Meet service uptime.
 - SendGrid/Email service reliability for transactional emails.
 - MongoDB Atlas cloud backup service.

Chapter 3

External Interface Requirements

3.1 User Interfaces

Public Website

- **Theme:** Dark luxury (#080808 background, gold #C9A84C accents).
- **Typography:** Playfair Display (headings), Montserrat (body).
- **Animations:** Market ticker (continuous left-scroll, pause on hover), card reveal on scroll.
- **Navigation:** Sticky navbar, hamburger menu on mobile.

Student Dashboard

- **Layout:** Sidebar navigation (collapsible on mobile), top bar with user menu, main content area with card-based panels.
- **Components:** Progress bars (gold fill), collapsible phase cards, embedded YouTube iframes (16:9), schedule calendar.

Admin/Super Admin Dashboards

- **Data Tables:** Striped rows, sortable columns, search/filter bars, export to CSV buttons.
- **Stats Cards:** Large Playfair Display numerals, gold color, trend indicators.
- **Destructive Actions:** Confirmation dialogs for delete/suspend/role change.

Refer to Section 4 of the original SRS for detailed UI specifications (UI-01 to UI-07).

3.2 Hardware Interfaces

No direct hardware interfaces. The system interacts with client devices (desktops, tablets, mobiles) via standard web protocols.

3.3 Software Interfaces

Software Component	Purpose	Interface Details
Cashfree API	Payment processing	REST API, webhook POST to <code>/api/webhooks/cashfree</code> , signature verification.
Google Meet	Live sessions	Admin provides link; system stores and reveals to student 15 min before start.
YouTube Data API (indirect)	Video embedding	Iframe embed using unlisted URL stored in DB.
SendGrid	Email delivery	REST API for sending verification, invoice, reminder emails.
WhatsApp Business	Chat	<code>wa.me/</code> links with pre-filled text messages.
AWS S3	File storage	PDF invoices, course PDFs, trade audit images.

3.4 Communications Interfaces

- **Protocol:** HTTPS only (HTTP redirects to HTTPS).
- **API Format:** REST over JSON.
- **Authentication:** JWT tokens in `httpOnly` cookies.
- **Webhook:** Cashfree webhook uses POST with `X-Webhook-Signature` header for verification.
- **Email:** SMTP or SendGrid API.
- **Rate Limiting:** 100 requests/min per IP on public routes; 5 failed logins lock for 15 min.

Chapter 4

System Features

4.1 System Feature 1: Authentication & User Management

4.1.1 Description and Priority

Priority: Critical. Enables secure registration, login, and role-based access for students, admins, and super admin.

4.1.2 Stimulus/Response Sequences

- **Register:** User submits form → System validates → Sends verification email → Redirects to pricing.
- **Login:** User submits credentials → System validates → Issues JWT cookies → Redirects to dashboard (student) or admin panel.
- **Forgot Password:** User requests reset → System sends email link → User sets new password.

4.1.3 Functional Requirements

- **FR-06:** Student Registration with fields: Full Name, WhatsApp Number (10-digit), Email (unique), Password (min 8 chars, 1 uppercase, 1 number), Interest, Terms agreement.
- **FR-07:** Student Login via Email/Password. JWT access token (15 min) and refresh token (7 days) stored in `httpOnly` cookies. "Remember Me" extends refresh token to 30 days.
- **FR-08:** Admin Login at `/admin/login`. 2FA via email OTP (Phase 1). Session timeout: 8 hours inactivity.
- **FR-09:** Super Admin Login at `/superadmin/login`. Mandatory 2FA via email OTP. IP whitelisting optional.

4.2 System Feature 2: Payment Integration (Cash-free)

4.2.1 Description and Priority

Priority: Critical. Processes course enrolment fee of INR 39,999 (inclusive GST) via Cashfree.

4.2.2 Stimulus/Response Sequences

- **Payment Success:** User pays → Cashfree webhook hits `/api/webhooks/cashfree` → System verifies signature → Updates student status from `pending_payment` to `active` → Sends welcome email + invoice PDF → Unlocks dashboard.
- **Payment Failure:** Webhook notifies failure → System logs failure → User sees retry option.

4.2.3 Functional Requirements

- **FR-10:** Cashfree Checkout. Order ID format: `RMF-{YYYY}-{8-char alphanumeric}`. Supported methods: UPI, Card, Net Banking, EMI, Wallets.
- **FR-11:** Invoice Generation. Auto-generates GST-compliant PDF invoice on success. Emailed and stored. Downloadable from dashboard.
- **Refund Policy:** No refunds per T&C. Manual exceptions via Super Admin panel.

4.3 System Feature 3: Student Dashboard (Post-Payment)

4.3.1 Description and Priority

Priority: Critical. Provides authenticated, paying students access to course content, live classes, support, and trade audits.

4.3.2 Stimulus/Response Sequences

- **View Course Phase:** Student clicks phase → System checks unlock status → Displays materials (PDFs, YouTube embeds).
- **Join Live Class:** Student clicks "Join Now" (active 15 min before session) → System opens Google Meet link in new tab.
- **Submit Trade Audit:** Student uploads screenshot + notes → Admin reviews → Student receives email notification + feedback.

4.3.3 Functional Requirements

- **FR-12:** Dashboard Home with welcome banner, progress bar, upcoming session countdown, announcements.
- **FR-13:** Course Phases (5 phases: Foundation, Technical, Advanced, Risk, Execution). Sequential unlock (admin override). Materials: PDFs, notes, YouTube unlisted videos. "Mark Complete" per phase.
- **FR-14:** Live Classes (Google Meet). Schedule display, Join button (15-min window), session history. No replay in Phase 1.
- **FR-15:** My Profile. Edit name/phone, change password, download invoice, language preference (English/Hindi).
- **FR-16:** Support Ticket System. Raise ticket (Subject, Category, Message). Auto ticket number RMF-TKT-XXXXX. Status: Open/In Progress/Resolved. Email notifications.
- **FR-17:** Announcements. View admin announcements; unread badge on nav.
- **FR-18:** Trade Audit Submission. Form: Date, Pair, Direction, Entry/Exit Price, P/L, Notes + screenshot upload. Admin feedback via email.

4.4 System Feature 4: Admin Dashboard (Mentors)

4.4.1 Description and Priority

Priority: High. Enables Vishal and Nilesh to manage students, content, live sessions, and support.

4.4.2 Stimulus/Response Sequences

- **Schedule Live Session:** Admin fills form (Date, Time, Topic, Mentor, Google Meet Link) → System saves to DB → Students see in their dashboard.
- **Reply to Ticket:** Admin types reply → System sends email to student and updates ticket status.

4.4.3 Functional Requirements

- **FR-19:** Admin Home with stats: Total Students, Revenue MTD, Pending Queries, Upcoming Sessions.
- **FR-20:** Student Management. Table view (Name, Email, Phone, Status, Progress). Search/filter. Manual activate/suspend. Export CSV.
- **FR-21:** Course Content Management. Upload PDFs/notes per phase. Add YouTube unlisted URL. Lock/unlock phases per student or globally.
- **FR-22:** Live Session Management. Schedule/edit/cancel sessions. Students notified on cancellation. "Join" button activates 15 min before.

- **FR-23:** Announcement Management. Create with tag (General/Course/Live/Important). Email blast for "Important".
- **FR-24:** Support Ticket Management. View open tickets. Reply and change status (Open → In Progress → Resolved).
- **FR-25:** Trade Audit Review. List pending submissions. View screenshots. Post written feedback. Mark reviewed → email student.

4.5 System Feature 5: Super Admin Dashboard

4.5.1 Description and Priority

Priority: Critical (for platform owner). Provides complete visibility and control over the entire system.

4.5.2 Stimulus/Response Sequences

- **View Revenue:** Super Admin loads dashboard → System aggregates all successful payments, shows chart and trend.
- **Promote User to Admin:** Super Admin selects user → Changes role → System updates role and logs action.

4.5.3 Functional Requirements

- **FR-26:** Super Admin Home. Metrics: Total Revenue (all-time/MTD), Total Students (active/pending/suspended), System Health (API/DB/Webhook), Recent Admin Activity (last 20 actions).
- **FR-27:** Full User Management. View all users (including admins). Create new admin accounts. Reset passwords. Delete users with cascade. Promote/demote roles (except Super Admin).
- **FR-28:** Payment & Revenue Management. Full transaction history. Filter by date/status/method. Manual payment entry. Export reports (CSV/Excel/PDF). Initiate manual refund.
- **FR-29:** Query Resolution Center. View all support tickets. Reassign tickets between admins. Add internal notes (not visible to student). Generate support reports.
- **FR-30:** Platform Settings. Update site-wide settings (WhatsApp numbers, email config). Toggle maintenance mode. Manage email templates. View error logs.
- **FR-31:** Admin Activity Audit Log. Logs every admin action with timestamp, admin name, action, IP, affected entity. Super Admin can search/filter/export logs.

Chapter 5

Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-ID	Requirement	Target
NFR-01	Page Load Time (FCP)	≤ 2 seconds on 4G
NFR-02	API Response Time (95th percentile)	≤ 500 ms
NFR-03	Dashboard Load Time	≤ 3 seconds
NFR-04	Payment Checkout Redirect	≤ 2 seconds to Cashfree
NFR-05	Concurrent Users	Handle 500 concurrent users
NFR-06	Database Query Performance	≤ 100 ms (indexed queries)

5.2 Safety Requirements

The system does not involve physical safety-critical functions. However, data integrity is critical: payment webhooks must be idempotent to avoid double activation. Automated daily MongoDB backups with 30-day retention are required.

5.3 Security Requirements

NFR-ID	Requirement
NFR-07	All API endpoints require authentication except public routes.
NFR-08	Role-based access control (RBAC) enforced server-side.
NFR-09	Passwords hashed with bcrypt (min 12 rounds).
NFR-10	JWT stored in httpOnly, Secure, SameSite=Strict cookies.
NFR-11	HTTPS enforced.

NFR-12	Cashfree webhook signature verification on every webhook.
NFR-13	Rate limiting: 100 req/min per IP; 5 failed logins = 15-min lock-out.
NFR-14	Input sanitization and validation on all fields (server-side).
NFR-15	CORS only allow rajmudrafinetech.com .
NFR-16	No sensitive data in frontend logs.
NFR-17	MongoDB: no raw queries; Mongoose schema validation.
NFR-18	Admin audit log stored separately and cannot be deleted by admin.

5.4 Software Quality Attributes

- **Usability:** WCAG 2.1 AA minimum (contrast ratio 4.5:1). Keyboard navigable. Clear error messages.
- **Reliability:** 99.5% monthly uptime SLA. Error monitoring via Sentry.
- **Maintainability:** Code split by route, lazy loading for dashboard modules. Environment variables for all secrets.
- **Localizability:** Bilingual support (English + Hindi) with language switcher in student dashboard.

5.5 Business Rules

- **BR-01:** A student can only access course content after payment status becomes 'active'.
- **BR-02:** Course phases unlock sequentially unless an admin overrides.
- **BR-03:** Google Meet link is only visible to students between 15 minutes before start time and scheduled end time.
- **BR-04:** No refunds are processed automatically. Only Super Admin can manually initiate a refund via Cashfree dashboard.
- **BR-05:** Admin actions (excluding view) must be logged in the audit log.
- **BR-06:** The platform price (INR 39,999) is configurable only by Super Admin and requires confirmation.

Chapter 6

Other Requirements

6.1 Database Requirements

- MongoDB Atlas (M10+ cluster) for production.
- Collections: `users`, `courses`, `payments`, `sessions`, `tickets`, `announcements`, `tradeaudits`, `auditlogs`, `notifications`.
- Indexes on: `users.email`, `payments.orderId`, `sessions.startTime`.
- Automated daily backups with 30-day retention.

6.2 Legal Requirements

- Compliant with Indian IT Act and RBI guidelines for payment aggregation.
- Privacy Policy and Terms & Conditions must be accepted during registration.
- GST-compliant invoices must be generated for every successful payment.
- No refund policy must be clearly displayed on the pricing page.

6.3 Internationalization

- Phase 1: English only.
- Phase 2: Hindi language support. Language switcher in student dashboard. Key UI strings, email templates, and error messages translated.

Appendix A

Glossary

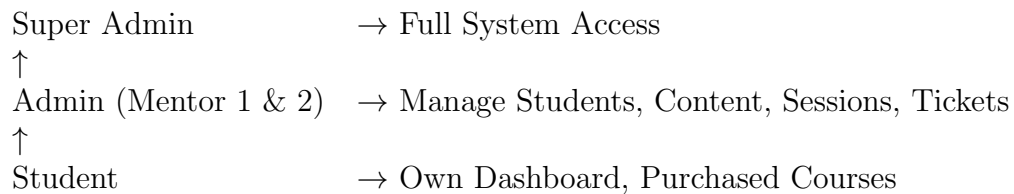
Term	Definition
MERN	MongoDB, Express.js, React.js, Node.js — full-stack JavaScript framework.
JWT	JSON Web Token — stateless authentication mechanism.
Cashfree	Indian payment gateway supporting UPI, cards, net banking, EMI, wallets.
Google Meet	Video conferencing tool used for live trading sessions.
YouTube Un-listed	YouTube video accessible only via direct link — not searchable publicly.
Webhook	HTTP callback triggered by Cashfree to notify payment status changes.
RBAC	Role-Based Access Control — permission model based on user roles.
IST	Indian Standard Time — UTC+5:30.
VTP	Vishal Tompe Patil — Founder & Mentor, Rajmudra Fintech.
NTP	Nilesh Thore Patil — Founder & Mentor, Rajmudra Fintech.
Trade Audit	Review service where mentors provide feedback on student's actual trades.
Super Admin	Platform owner with unrestricted system access.

Appendix B

Analysis Models

User Role Hierarchy Diagram

Figure B.1: User Roles and Access Levels



Payment Flow Sequence

1. Student clicks "Enrol Now" → Creates order in DB.
2. Redirect to Cashfree checkout.
3. Student completes payment.
4. Cashfree sends webhook to `/api/webhooks/cashfree`.
5. Server verifies signature.
6. If success: Update user status to 'active', generate invoice, send welcome email, unlock dashboard.
7. If failure: Log failure, notify student to retry.

Appendix C

To Be Determined List

- **TBD-01:** Real-time API integration for live market ticker (Forex/stock data). Planned for Phase 2.
- **TBD-02:** Google Calendar API integration for automatic Meet link generation. Planned for Phase 2.
- **TBD-03:** Automatic attendance tracking for live sessions. Planned for Phase 2.
- **TBD-04:** WhatsApp confirmation message on registration. Optional, Phase 2.
- **TBD-05:** 2FA via mobile OTP (currently email OTP). Planned for Phase 2.